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Statement of Authorship

I hereby declare that this document has been composed by myself and describes my own work, unless otherwise acknowledged in the text.

Passau, August 19, 2025

Abstract

A short summary of what is going on here.

Deutsche Zusammenfassung

Kurze Inhaltsangabe auf deutsch.

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1. Introduction

This chapter should contain

1. A short description of the thesis topic and its background.
2. An overview of related work in this field.
3. Contributions of the thesis.
4. Outline of the thesis.

2. Preliminaries

Before tackling the problem, we will introduce some concepts that are necessary for the understanding of this problem.

Definition 2.1 (Graph Embedding)**.** *a Graph embedding of a graph G on a*

Definition 2.2 (Level Assignement)**.** *A level assignement l of a level graph G is a mapping of vertices of G to levels.*

$$l : V(G) \rightarrow \mathbb{N}$$

Definition 2.3 (Proper Level Graph)**.** *a Proper Level Graph G is a graph with a level assignement l that defines in which level every vertex is while respecting that for every edge $(u, v) \in E(G)\ |\, l(u) - l(v)\, | = 1$*

Definition 2.4 (Planar Graph)**.** *a Planar Graph G is a graph that can have an embbeding on a 2-dimensional Euclidean space $\mathcal{R}$ in a way that edges only intersect with each other at their endpoints.*

2.0.1. Equivalence relation and the 2-SAT formulation

Randerath et al. [RSB$^+$01] present some simple constraints that decide the existence of a level-planar drawing. The constraints are as follows:

- *consistency:* $uw \iff \neg wu \quad \forall\, uw \in \mathcal{V}$
- *transitivity:* $(uw \wedge wx) \Rightarrow ux \quad \forall\, uw, wx \in \mathcal{V}$ and with $ux \in \mathcal{V}$
- *planarity:* $uw \iff vx \quad \forall uw, vx \in \mathcal{V}$ with $(u, v), (w, x)$ ($\in E$ and are independant)

Definition 2.5 (Equivalence relation)**.** *An equivalence relation on a set X is a binary relation $\sim$ on X satisfying the following constraints:*

- *reflexivity:* $a \sim a \forall a \in X$.
- *symmetry:* $a \sim b$ *implies* $b \sim a \quad \forall a, b \in X$.
- *transitivity:* $a \sim b \wedge b \sim c$ *implies* $a \sim c \quad \forall a, b, c \in X$.

The planarity constraint can be formulated as an *equivalence relation* over the set $\mathcal{V}$ according to Randerath et al. [RSB+01].

Definition 2.6 (Orbit relation)**.**
Let's consider a proper level graph $G = (V, E)$*, and a boolean set* $\mathcal{V} = \{uw \mid u, w \in V,\, l(u) = l(w)\}$*. The orbit relation is an equivalence relation on the boolean set* $\mathcal{V}$ *with a binary relation* $\sim$ *on* $\mathcal{V}$ *satisfying the following constraints:*

- *reflexivity:* $uw \sim uw \quad \forall\, uw \in \mathcal{V}$
- *symmetry:* $uw \sim vx$ *implies* $vx \sim uw \quad \forall\, uw, vx \in \mathcal{V}$
- *transitivity:* $(uw \sim zy) \wedge (zy \sim vx)$ *implies* $uw \sim vx \quad \forall\, uw, zy, vx \in \mathcal{V}$

The interperation of the orbit of a variable $uw \in \mathcal{V}$ is that for each boolean variable vx of $\mathcal{V}$ in the orbit of uw we have: $(u < w) \iff (v < x)$.[RSB+01]

Theorem 2.7. *A proper level graph* G *is level planar iff there exists no* $uw \in \mathcal{V}$ *such that* wu *in orbit of* uw*.*

Proof. Refer to Theorem 6 from [RSB+01]. □

by considering the orbit which represents the *planarity* constraint and by considering the *consistency* constraint, that is necessary to prove the existence of a level planar based on the Theorem 2.7. Randert et al [RSB+01] proves that the transitivity clause is not needed for the satisfiability of this system of constraints and it is equivalent to the satisfiability of a reduced constraint system which omits the transitivity clause.

The reduced constraint system can be expressed in terms of 2-SAT, which can be solved efficiently with a quadratic-time algorithm for testing level planarity [Kro67]. Randert et al. also prove that a solution can be made transitive without invalidating the other constraints in Theorem 2.7.

2.0.2. Hanani-Tutte drawing and Embedding

Theorem 2.8. *If* G *has an (x-) monotone drawing with every pair of independant edges crosses evenly then* G *has (x-) monotone embedding with the same vertex locations (over x).*

Proof. Refer to theorem 3.1 from [FPSŠ12] □

2.0.3. Equivalence classes of the orbits and Hanani-Tutte drawing

In this section we want to link the equivalence classes with the Hanani-Tutte drawing induced from the truth assignement of the 2-SAT formulation that is based on equivalence classes of the orbits.

Definition 2.9 (Equivalence class)**.** *For a given set* X *and an equivalent relation* $\sim$ *in* X*, for an element* x *of* X*, the equivalence class* $[x]_\sim = \{y \sim x, y \in X\}$*.*

The set of all equivalence classes of X *with a relation* $\sim$ *is represented with* $X/\sim$

Definition 2.10 (Orbit class)**.** *For a boolean variable* $uw \in \mathcal{V}$*, the orbit class* $[uw]_\sim$ *is the set of all elements* $vx \in \mathcal{V}$ *with* $vx \sim uw$*.*

The set of all orbit classes of $\mathcal{V}$ *with the relation* $\sim$ *is* $\mathcal{V}/\sim$*.*

Lemma 2.11. $\forall uw \in \mathcal{V}\ \forall vx \in [uw]_\sim : \neg vx \in [\neg uw]_\sim$

Proof. by the consistency clause, $\forall uw \in \mathcal{V} : uw \iff \neg wu$ and $\forall vx \in [uw]_\sim vx \iff uw$ therefore

$$\forall uw \in \mathcal{V}\ \forall vx \in [uw]_\sim vx \iff \neg wu$$

thus

$$\forall uw \in \mathcal{V}\ \forall vx \in [uw]_\sim \neg vx \iff wu \iff \neg uw$$

and that's why

$$\forall uw \in \mathcal{V}\ \forall vx \in [uw]_\sim \neg vx \in [\neg uw]_\sim$$

□

Each assignment of the equivalence classes of the orbit is represented in the hanani-tutte drawing Γ^* of G^* in the level l where the single vertex v' of origin v in G is one of the pair of vertices in the equivalence class and the other vertex w of the pair is the edge e' passing through the level l with an endpoint of origin w in the graph G. If for example we assign one to $[vw]_\sim$, then the vertex v' lays before e' on the level l in the drawing Γ^*.

Definition 2.12 (Horizental Gang). *a Horizental Gang is a set of distinct vertices that belong to the same level of the level graph G and that are the distinct endpoints of adjacent edges.*

Definition 2.13 (Base Orbit Class). *a base orbit class $\mathcal{B}([uv]_\sim)$ is the merge of $[uv]_\sim$ and $[vu]_\sim$. The orbit classes $[uv]_\sim$ and $[vu]_\sim$ are the directions of the base orbit class $\mathcal{B}([uv]_\sim)$.*

Lemma 2.14. *Let a level planar graph $G = (V, E)$ with a level assignment l and let the drawing Γ^* be the induced Hanani-Tutte drawing of the graph G^* induced from the orbit classes of G. An odd crossing in Γ^* is equivalent to a cyclic relation between a triple of members of a horizental gang V_{HG}, in other words a pair or triple of base orbit classes of members of the horizental gang can have directions that are cyclic.*

Proof. TODO proof □

The algorithm that we designed might merges some of the orbit classes, we want to seperate the orbit classes before the execution of the algorithm from any other state of the $\mathcal{V}/\sim$, for this reason we introduce this definition.

Definition 2.15 (first-gen orbit class). *A first-gen orbit class is an orbit class purely based on the planarity constraint, wich is not contaminated with other elements of $\mathcal{V}$ that are not par of it from the planarity constraint.*

3. Transitivity constraint and Merging of the equivalent classes

After linking the Equivalent Classes to the hanani-tutte drawing induced by the 2-Sat formula (insert paper), and also presenting the cases that lead to a cyclic relation between some vertices of the same level, we can now present our algorithm that reduce the set equivalent classes resulted from resolving the 2-SAT formula (insert papers).

3.1. Algorithm description

*TODO THIS NEEDS REFACTORING

We have as input a planar proper level graph $G = (V, E, l)$ with a level assignement l, and equivalence classes based on the 2-SAT formulation of level planar embedding presented by Randert et al [RSB$^+$01]. The 2-SAT formulation tests wether a proper level graph is planar. The truth assignements that solve the 2-SAT formula give embeddings that respect the planarity constraints but some of the assigments can't be translated to an embeddings in a plane surface due to the existence of cyclic relation between some of the vertices from the same level. These set of these assignements is equal to the set of possible combinations of assignement of the equivalence classes.

The goal of the algorithm is to merge some of the equivalence classes which means less possible truth assigment claiming that with the newly merged equivalence classes all the possible assignements of these equivalence classes lead to an embedding.

We first make the edges of G point to the same direction.(part of the computation of the equivalent classes)

We traverse the graph level by level, we also keep track of the vertices that we went throught by adding them to a graph G' that is initially an empty graph. Let h be the current level. In each level we go throught all the vertices, for each vertex v we execute two main taskes. The first task is to merge the equivalence classes of neighboring vertices to v that belong in the same adjacent level to level h. The second task is to merge the equivalence classes of pair of vertices (u, w) with u and w are in different connected components C_1 and C_2 of $G - \{v\})$ and v is a cut vertex of G connecting C_1 and C_2.

For the first task, we consider vertices in $V_{\mathrm{v}}^{(h-1)}$ adjacent to the vertex v on the level $h-1$. We compute an ordering $\mathcal{V}$ of the vertices in $V_{\mathrm{v}}^{(h-1)}$ so that for all pairs of vertices $(u, w) \in V_{\mathrm{v}}^{(h-1)} \times V_{\mathrm{v}}^{(h-1)}$ and $u < w$ the equivalence classes point to the same direction.

Algorithm 3.1: MergeAdjacentEdgestOrbitClasses

```
Input: proper level Graph G = (V, E, l) and Eq
Output: Eq
/* reducing the orbit classes of adjacent edges                  */
forall h ∈ l(V) do
    forall v ∈ {V | l(v) = l(h)} do
        visited ← null
        parentEq ← null
        reverse ← false
        forall w ∈ adj(h,v) do
            member ← visited
            while member.next ≠ null do
                if parentEq = null then
                    parentEq ← eq(u, w)
                if ((w, u) ∈ parentEq) then
                    reverse ← true
                    member.previous ←
                      {data: w, perivous: member.previous, next: member}
                if (eq(u, w) ∉ parentEq) ∧ (eq(w, u) ∉ parentEq) then
                    if reverse then
                        Merge eq(w, u) with parentEq
                    else
                        Merge eq(u, w) with parentEq
                member.next ← {data: w, perivous: member, next: null}
            if visited = null then
                visited ← {data: w, perivous: null, next: null}
```

We merge those equivalence classes of the pairs $(u, w) \in \mathcal{V}$ with $u < w$. We merge also the equivalence classes of the pairs $(u, w) \in \mathcal{V}$ with $u > w$ to hold the consistency clause.

We will also do the same steps for vertices in $V_{\mathrm{v}}^{(h+1)}$ that are adjacent to the vertex v on the level $h + 1$.

For the second task. we add v to G'. If the vertex v is a cut-vertex in G', we look for the connected components $S_{\mathcal{C}}$ in $G' - \{v\}$. We pick an arbitrary absolute ordering $\mathcal{V}_{S_{\mathcal{C}}}$ of the connected components in $S_{\mathcal{C}}$. For each pair of connected components $(C_1, C_2) \in \mathcal{V}_{S_{\mathcal{C}}}$, for each $w_1 \in C_1$ with $l(w_1) = h - 1$ we merge the equivalence classes of w_1 with all $w_2 \in C_2$ and $l(w_2) = h - 1$.

3.1.1. Merging orbit classes of horizental gangs

We present the Algorithm 3.1.

3.2. Correctness of the algorithm

3.2.1. Merging orbit classes of horizental gangs

In this section, we will prove that the part of algorithm, that might merges some of the orbit classes between members of the same horizental gangs, leads to a new set of orbit classes that for all their assignements there exists a weak Hanani-Tutte drawing in G^*.

We will define two invariants that we will use in some of our proofs.

- The invariant H_1 of having the orbit classes not breaking the *consistency constraint* and the *planarity constraint*
- The invariant H_2 of having acyclic relations between the members of the horizental gang of the set.

Lemma 3.1. *let* $(w_1, \ldots, w_j)$ *subset* HG_A *of vertices of a horizental gang* HG*, that respect having the orbit classes not breaking the consistency constraint and the planarity constraint and that there is no assignement of the orbit classes in* HG_A *resulting in a cylic relation. let* $w_{j+1} \in HG$ *After the algorithm 3.1 treats* w_{j+1} *in the loop.* $HG_A \cup \{w_{j+1}\}$ *also respects same constraints.*

Proof. To prove this we will prove by induction that merging the equivalence class between w_{j+1} and the members of HG_A hold the two constraints. Let VHG the set of visited members of the horizental gank HG^j_A.

For the base case we have only one visited member $\mathrm{VHG}_1 = \{w_1\}$ from the gang HG^j_A. Which means we are considering only one orbit class $eq\,(w_1, w_{j+1})$, which doesn't lead to any cyclic relation since we need three vertices at least for that to be possible.

For the induction step, let $\mathrm{VHG}_k = (w_1, \ldots, w_k)$ with $k < j$. let's consider the orbit class $eq\,(w_k, w_{j+1})$. If the base orbit class $\mathcal{B}\,(w_k, w_{j+1})$ is different than the base of the root equivalence class then we merge the corresponding direction of the base orbit class $\mathcal{B}\,(eq\,(w_k, w_{j+1}))$, depending on the mode, with the root equivalence class $eq\,(w_0, w_1)$.

This first keeps the invariant H_1 because merging $eq\,(w_k, w_{j+1})$ and the root equivalence class, without losing generality, is only possible if the opposite direction of $eq\,(w_k, w_{j+1})$ which is $eq\,(w_{j+1}, w_k)$ doesn't exist in the parent equivalence class, which means the consistency clause is still respected, and since we are just merging orbit classes and not adding any new relations between vertices and the consistency is respected, the planarity constraint is also respected. For the invariant H_2, let's prove by contradiction that merging the base orbit class of w_k and w_{j+1}, with a direction depending on the phase of the algorithm, with the parent equivalence class will lead to a cyclic relation in one of the 'new' triples (x, w_k, w_{j+1}) with $x \in \mathrm{VHG}_k - \{w_k\}$. For the first mode, where $inverse = false$ which means that we didn't find a member of the gang that must be after w_{j+1}. That can't happen if the base of orbit class (w_k, w_{j+1}) is not part of the base of the parent orbit class, because whatever the direction of the base of the orbit class $\mathcal{B}\,((w_k, w_{j+1}))$, it doesn't lead to a cyclic relation if we merge the base of that orbit class with one of the direction with the parent orbit class.

For the case where we only have the parent orbit class between the triple. It can't be an first-gen equivalence class as these don't lead to a cyclic relation (TODO lemma?). It the the orbit class in the triple is a result of one or multiple merges, We know that a triple of gang members that can have a cyclic relation don't share more than one edge with any other triple of gang members (TODO lemma) which means that, we can't have this case since if we had at least one merge of the orbit class in the triple, it only means at least two equivalence classes were part of another triple of gang members. For the second mode, where $inverse = true$ where we found that w_{j+1} needs to be before a gang member of VHG. For each triple (x, w_k, w_{j+1}) with $x \in \mathrm{VHG}_k - \{w_k\}$, they can induce two orbit classes or one. In the case of two, the base orbit class between x, w_{j+1} is of the direction from w_{j+1} to x and the base orbit class x, w_k is of a direction from x to w_k since x comes before w_k in visited linked list. Since we are in the reverse mode of the algorithm, the algorithm will merge the base orbit class of w_k, w_{j+1} with the direction from w_{j+1} to w_k

with the other mentionned induced orbit classes, which contradicts. For the case, where we have only one orbit class induced from the triple, it's not possible due to the same contradiction from the first mode.

The merge of the appropriate orbit class from the base of (w_{j+1}, w_k) doesn't affect any existing triple order, since we are just changing the new orbit class.

Therefore merging the equivalence classes between w_{j+1} and the members of HG_A^j, when needed, hold the two invariants.

Thus adding the vertex member w_{j+1} to the gang HG_A^j will hold both invariants.

□

Theorem 3.2. *The algorithm 3.1 is correct and hold the invariant H_1 while removing all the possible cyclic relations between members of the same horizental gang.*

Proof. We can proof the correctness of this algorithm 3.1 by proving that the resulted orbit classes of the algorithm have no assignement that for a triple or pair of base orbit classes $\mathcal{B}(\mathcal{V}_{HG})$ of members of a horizental gang we can find a combinaison of directions that lead to a cyclic relation between orbit classes from the triple or pair of base orbit classes $\mathcal{B}(\mathcal{V}_{HG})$. By Theorem 2.8, we can prove it by induction on odd pairs of the Hanani-Tutte drawing induced by a truth assignement. And by Lemma 2.14 We can prove it by induction on the horizental gangs and removing any possible combinaison of directions between base orbits of members of thet horizental gang.

We want also to hold an invariant H_1 of having the orbit classes not breaking the *consistency constraint* and the *planarity constraint* which also mean that any combination assignements of the orbit classes can be drawn as a hanani-tutte drawing.

The base case is our *first-gen* orbit classes of the graph G induced from the 2-SAT formulation. all the possible combinations of assignements of the first-gen orbit classes can be lead to a hanani-tutte drawings [BRS22].

For the induction step, let's consider that the possible combinations of assignements of the set of the orbit classes Or_i of graph G after treating i sets of horizantal gangs, holds the invariant H_1

We want to prove that by merging some orbit classes correspending to the horizantal gang HG_{i+1} resulting in the set of equivalence classes Eq_{i+1} the invariant H_1 holds and also taht we removed any possilbe cyclic relation between any triple of the horizental gang HG_{i+1}.

let's consider $(w_1, \ldots, w_l)$ the vertices of the horizantal gang HG_{i+1}.

to prove that the merging done in the algorithm is correct, we need to prove that the merging resulting in Eq_{i+1} holds the invariant of H_1 and also hold the invariant H_2 of having acyclic relations between the members of the horizental gang HG_{i+1}.

We proceed the proof by induction on members of the horizental gang HG_{i+1} which correspendence to the loop of the gang in the algorithm.

The base case consists only from a single vertex member of the horizental gang HG_{i+1}^1 which holds the invariant of the induction on the horizental gangs and also hold the invariant of having acyclic relations between the members of the horizental gang HG_{i+1}^1 since the gang has no cyclic relations, since there are no relations.

For the induction step, let's consider that the provisional horizental gang HG_{i+1}^j consists of $(w_1, \ldots, w_j)$ members, with $j < l$, we want to prove adding the vertex member w_{j+1} to the gang HG_{i+1}^j will hold both invariants.

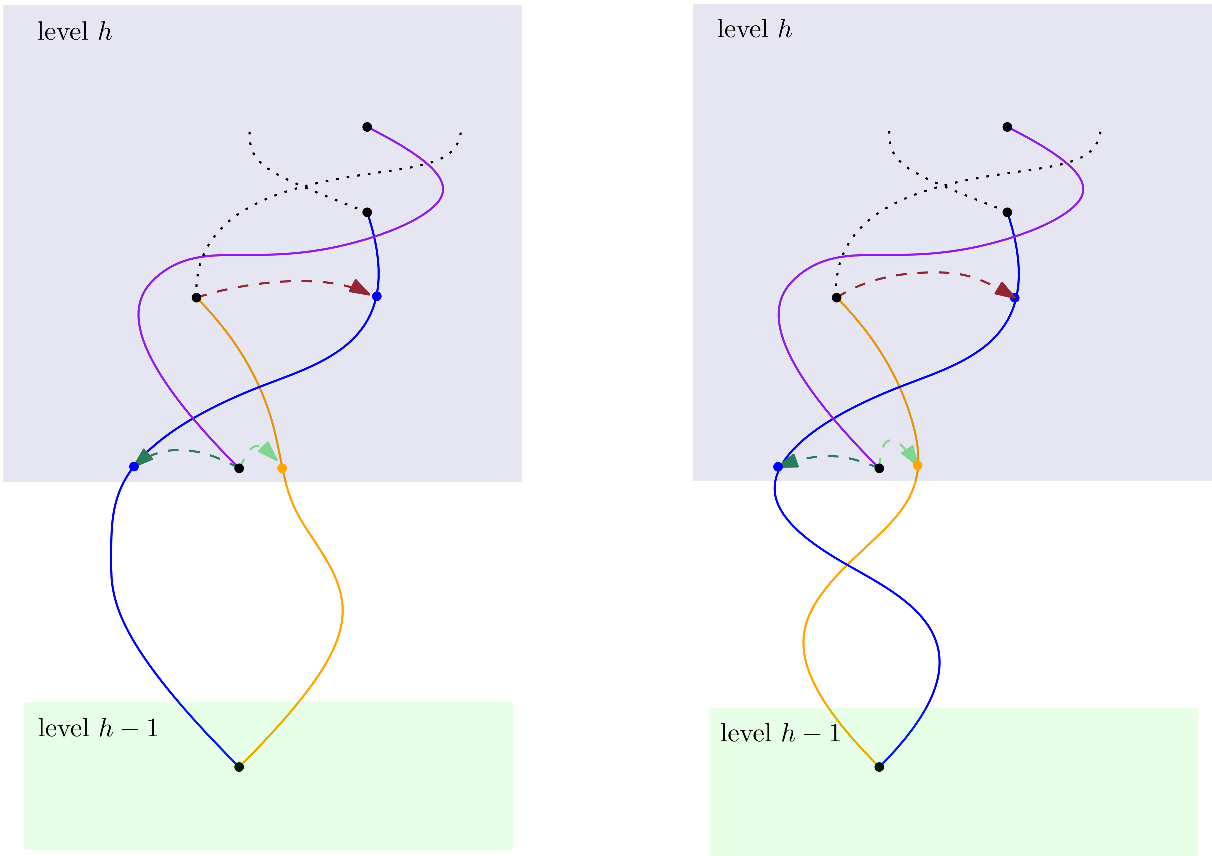

Figure 3.1.: consecutive edges in rotation that crosses oddly

By Lemma 3.1, we prove that adding the vertex w_{j+1} to the gang HG_{i+1}^{j} holds both invariants.

Which means that treating the horizental gang HG_{i+1} holds the invariant and also removes any possible cyclic relation between the horizental gang memberes of HG_{i+1} from the set of resulted orbit classes.

Therefore, we proved that the algorithm removes all the cyclic relations between any members of the same horizental gang. □

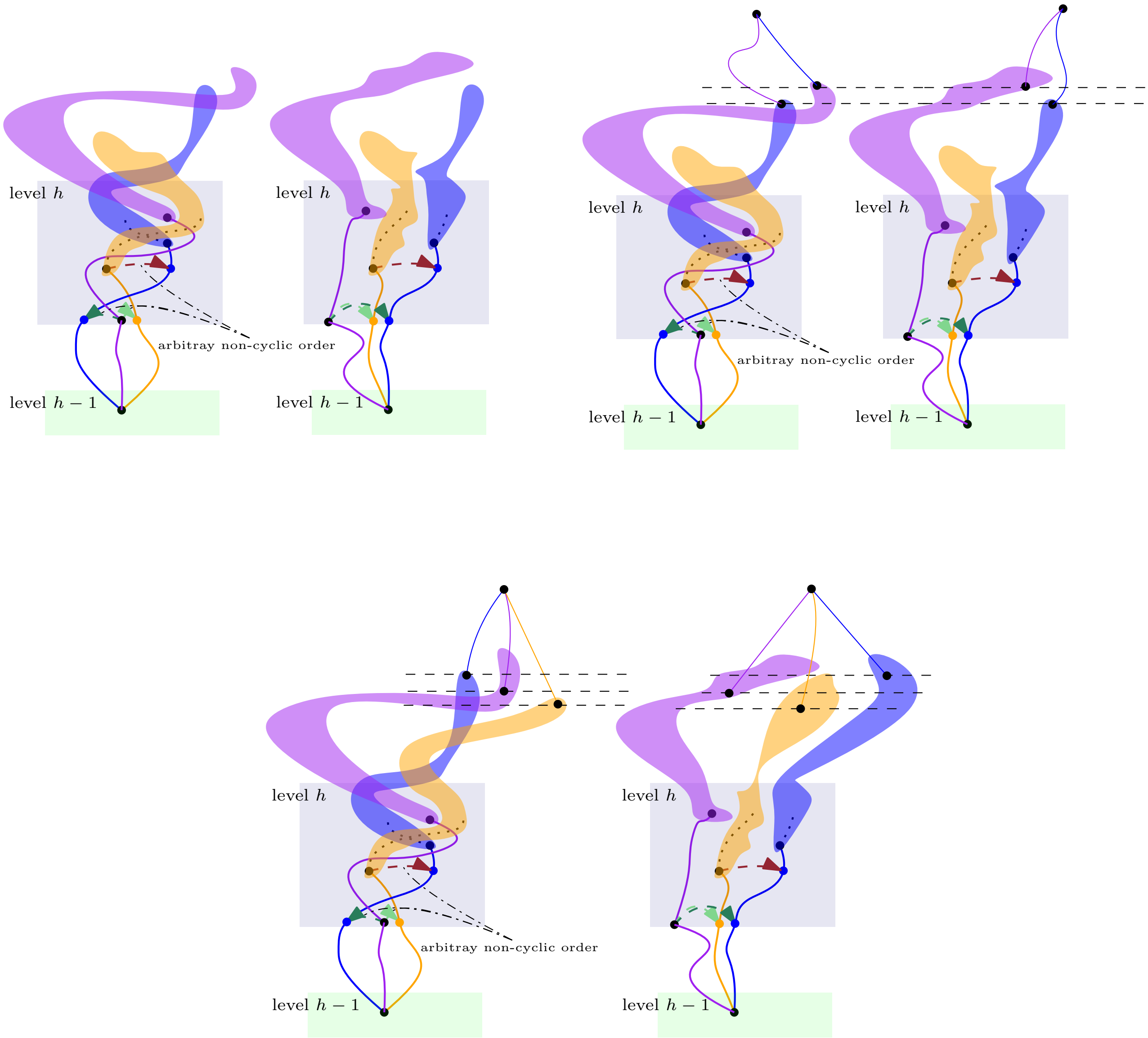

Figure 3.2.: in between even crossing edge

4. Conclusion

Summary and outlook.

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Appendix

A. Appendix Section 1

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Figure A.1.: A figure